MIL-F-60843(MU)
20 October 1967
SUPERSEDING
FRANKFORD ARSENAL
FA-PD-MI-2619, Rev. 0
28 June 1965

MILITARY SPECIFICATION

FIN, ASSEMBLY; M141

1. SCOPE

1.1 This specification is the applicable detail specification for Fin Assembly, M141.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue is effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified.

SPECIFICATIONS

Military
- MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

Federal
- Federal Test Method Standard No. 151 - Metals; Test Methods

Military
- MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
- MIL-STD-109 - Quality Assurance Terms and Definitions
- MIL-STD-1167 - Ammunition Data Card
- MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
- MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

DRAWINGS

Munitions Command
C-7549030 - Fin Assembly, M141
IEL-7549030 - Index to Inspection Equipment List for Fin Assembly, M141

PUBLICATIONS

ASA - B46.1 - Surface Texture, Surface Roughness, Waviness and Lay

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Cartridge housing.-The cartridge housing shall have an unobstructed primer flash hole as shown on the applicable drawings

3.3 Mechanical properties.-Mechanical properties shall be in accordance with the requirements of the applicable drawings and specifications.

3.4 Workmanship.-The requirements for workmanship are as shown on the applicable drawing, referenced specifications and the following:

3.4.1 Metal defective.-Soundness of the material shall be as required by the applicable material specifications.

3.4.2 Threads.-Threads shall be full and undamaged for the entire minimum length of depth as required by the applicable drawings and shall be in accordance with the provisions of Specification MIL-A-2550.

3.4.3 Burr.-No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

3.4.4 Foreign matter.- All parts and assemblies shall be free of chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 15 metal parts assemblies as follows:

10 Fin assemblies, complete
5 Sets of component parts, unassembled

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials including packaging and packing shall be the same as used in regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with Standard MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.2 Heat treatment lots.-Heat treatment lots shall contain cartridge housing forgings of the same alloy that have been heat treated in the same heat treating equipment at the same temperature in one unchanged process. The heat treatment lot thus constituted shall be the basis for selecting the sample for the tension test hereinafter prescribed. The maximum number of cartridge housing forgings in a heat treatment lot shall consist of 15,000 or fraction thereof.

4.3.1.3 Bar stock lot.-Cartridge housings machined from bar stock shall consist of not more than 15,000 machined from the same alloy, temper and size of bar stock.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ASA B46.1 shall be used as a basis for surface roughness determination. Four power (4X) magnification may be used in performing visual inspection for such defects as cracks, burrs, and foreign matter.

4.3.2.a Control tool manufacture.-Dimensions marked "control tool manufacture" shall be gaged at the beginning of production whenever tooling is changed or as a minimum as defined below.

(a) 90 degree + 30 degree angle-select and gage one housing from each machine at the start of each shift.
(b) Location of 20 flash holes.- Select and gage one housing from each machine every two hours.
(c) Size of 20 flash holes.- Select and gage one housing from each machine every two hours.
(d) Depth of primer flash hole counterbore (if applicable): Select and gage one housing from each machine every two hours.

1. All material produced prior to periodic inspection for items (a),(b) and (c) is acceptable; however, corrective action must be taken whenever a defective characteristic is found.

2. All material produced subsequent to last periodic inspection for item (d) will be rejected and screened whenever a defective characteristic is found and corrective action must be taken.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

4.3.2.1 Holder increment.- Drawing C7549026 covering a detail of Drawing C7549030.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Diameter, outside, major	Gage
202.	Diameter, outside, minor	Gage
203.	Diameter, inside	Gage
204.	Thickness	Gage
205.	Length of eye(s)	Visual (See 6.1)
206.	Width of eye(s)	Visual (See 6.1)
207.	Width of eye neck	Visual (See 6.1)
208.	Radii missing or incorrect	Visual (See 6.1)
209.	Surface finish improper	Visual (See 6.1)
210.	Eye(s) missing	Visual
211.	Metal defective	Visual
212.	Protective coating inadequate or damaged	Visual
213.	Burr	Visual
214.	Foreign matter	Visual

4.3.2.2 Housing, Cartridge, Drawing C7549028 covering a detail of Drawing C7549030.

CRITICAL		
	None defined	
MAJOR		
101.	Concentricity at front of taper with pitch diameter of body thread, including perpendicularity of front face	Gage
102.	Concentricity at rear of taper with pitch diameter of body thread, including perpendicularity at front face	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

4.3.2.2 (Cont'd)

Defect	Method of Inspection
MAJOR (Cont'd)	
103. Cartridge cavity not functionally concentric with pitch diameter of body thread	Gage
104. Primer flash hole not functionally concentric with cartridge cavity	Gage
105. Pitch diameter of body thread, max.	Gage
106. Minor diameter of body thread, max.	Gage
107. Pitch diameter of fin thread, min.	Gage
108. Major diameter of fin thread, min.	Gage
109. Diameter of cartridge cavity	Gage
110. Diameter of primer flash hole	Gage
111. Diameter of primer flash hole counterbore (if applicable)	Gage
112. Depth to bottom of body thread, cavity	Gage
113. Length of effective body thread, min.	Gage
114. Length of effective fin thread, min.	Gage
115. Flash hole(s) missing or excessive in number	Visual
116. Thread(s) damaged or not full	Visual
117. Metal defective	Visual
MINOR	
201. Pitch diameter of body thread, min.	Gage
202. Minor diameter of body thread, min.	Gage
203. Diameter at forward basic length	Gage
204. Diameter at rear basic length	Gage
205. Diameter of body thread counterbore	Gage
206. Depth of body thread counterbore	Gage
207. Depth to bottom of cartridge cavity	Gage
208. Length from rear face to fin seat	Gage
209. Length, total	Gage
210. Radii or chamfers missing or incorrect	Visual (See 6.1)
211. Surface finish improper	Visual (See 6.1)
212. Burr	Visual
213. Foreign matter	Visual

4.3.2.3 Fin Drawing F7549029 covering a detail of Drawing C7549030.

CRITICAL

None defined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

4.3.2.3 (Cont'd)

Defect		Method of Inspection
MAJOR		
101.	Concentricity of pitch diameter of primer thread with housing thread, including perpendicularity with front face of hub	Gage
102.	Primer thread counterbore (small) not functionally concentric with pitch diameter of primer thread	Gage
103.	Pitch diameter of housing thread, max.	Gage
104.	Minor diameter of housing thread, max.	Gage
105.	Pitch diameter of primer thread, max.	Gage
106.	Minor diameter of primer thread, max.	Gage
107.	Pitch diameter of through thread, max. (if applicable)	Gage
108.	Minor diameter of through thread, max. (if applicable)	Gage
109.	Diameter of housing thread counterbore	Gage
110.	Diameter of fin blades	Gage
111.	Diameter of primer thread counterbore (large)	Gage
112.	Diameter of primer thread counterbore (small)	Gage
113.	Depth of housing thread counterbore	Gage
114.	Depth of primer thread counterbore, (large)	Gage
115.	Depth of primer thread counterbore, (small)	Gage
116.	Length of effective housing thread, min.	Gage
117.	Length of effective primer thread, min.	Gage
118.	Thread(s) damaged or not full	Visual
119.	Metal defective	Visual
MINOR		
201.	Angle at front of fin blades	Gage
202.	Angle at rear of fin blades	Gage
203.	Angle between fin blades	Gage
204.	Pitch diameter of primer thread, min.	Gage
205.	Minor diameter of primer thread, min.	Gage
206.	Pitch diameter of through thread, min. (if applicable)	Gage
207.	Minor diameter of through thread, min. (if applicable)	Gage
208.	Diameter of hub, front	Gage
209.	Diameter of hub, parting line	Gage
210.	Diameter of hub, rear	Gage
211.	Diameter of skirt	Gage
212.	Length of skirt	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

4.3.2.3 (Cont'd)

Defect	Method of Inspection
MINOR (Cont'd)	
211. Length total	Gage
212. Thickness of skirt at front	Gage
213. Thickness of fin blades at rear	Gage
214. Radii or chamfers missing or incorrect	Visual
215. Surface finish, improper	Visual
216. Burr	Visual
217. Foreign matter	Visual

4.3.2.4 Fin Assembly, Drawing C7549030

CRITICAL	
1. Primer flash hole missing or obstructed	Visual
MAJOR	
101. Outside diameter of fin blades not functionally concentric with pitch diameter of body thread, including perpendicularity of front face of housing	Gage
102. Length, total, max.	Gage
103. Fin blades distorted or damaged	Visual
104. Increment holder missing or damaged	Visual
105. Thread(s) damaged or not full	Visual
106. Metal defective	Visual
MINOR	
201. Marking missing, incorrect or illegible	Visual
202. Burr	Visual
203. Foreign matter	Visual

4.3.2.5 Packed carton prior to sealing.

CRITICAL

None defined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

4.3.2.5 (Cont'd)

Defect	Method of Inspection
MAJOR	
None defined	
MINOR	
201. Separator(s) missing or damaged	Visual
202. Divider(s) missing or damaged	Visual
203. Improperly packed	Visual

4.3.2.6 Packed carton after sealing.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Cartons damaged (cut or damaged through all layers)	Visual
203. Carton not properly sealed	Visual
204. Marking missing, incorrect or illegible	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with 4.4.

4.3.3.1 Mechanical properties.

4.3.3.1.1 Hardness.

4.3.3.1.1.1 Heat treatment lots.- In order to determine uniformity of heat treatment, 80 cartridge housing forgings in the heat treatment lot (as defined by 4.3.1.2) shall be subjected to a hardness test with a Brinnell hardness tester or by any other method approved by the procuring activity.

4.3.3.1.1.2 Bar stock lot.- Eighty cartridge housings in the bar stock lot (as defined by 4.3.1.3) shall be subjected to a hardness test with a Brinnell hardness tester or by any other method approved by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

4.3.3.1.2 Tension test.

4.3.3.1.2.1 Heat treatment lots.-Two cartridge housings, one of the hardest and one of the softest as determined by 4.3.3.1.1.1 shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot.

4.3.3.1.2.2 Bar stock lots.-Two cartridge housings, one of the hardest and one of the softest, as determined by 4.3.3.1.1.2 shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot.

4.3.3.2 Torque test.-Five fin assemblies from each one hour's production or fraction thereof from each torque machine in operation shall be selected at random and tested to determine compliance with the applicable drawing. Failure of any fin assembly to comply shall be cause for rejection of the production represented by the sample.

4.3.4 Inspection equipment.-Index of Inspection Equipment Lists Number 7549030 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment designs shall be in accordance with the code designation on the applicable IEL.

4.4 Test methods and procedures.

4.4.1 Mechanical properties.

4.4.1.1 Hardness.

4.4.1.1.1 Heat treatment lots.-Prior to finish machining, the samples supplied in accordance with 4.3.3.1.1.1 will be subjected to a hardness test on either end of the cartridge housing. The test shall be performed using the equipment and procedures prescribed by Federal Test Method Standard No. 151. A minimum dwell time of 5 seconds shall be used.

4.4.1.1.2 Bar stock lots.-The samples supplied in accordance with 4.3.3.1.1.2 will be subjected to a hardness test on either end of the cartridge housing. The test shall be performed using the equipment and procedures prescribed by Federal Test Method Standard No. 151. A minimum dwell time of 5 seconds shall be used.

4.4.1.2 Tension test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

4.4.1.2.1 Yield strength and elongation.-From the cartridge housings selected in accordance with 4.3.3.1.2.1 and 4.3.3.1.2.2, one tension test specimen conforming to the largest obtainable Standard "R" type specimen as prescribed by Federal Test Method Standard No. 151 shall be taken longitudinally. The number of specimens is for the convenience of the contractor and Government and shall not restrict the contractor or Government from testing additional specimens if necessary to determine that the mechanical properties of the cartridge housing meet the drawing requirements. The criteria for performing the tests shall be as prescribed by Federal Test Method Standard No. 151.

4.4.2 Torque test.-The samples supplied in accordance with 4.3.3.2 shall be tested by applying the required assembly torque to the fin in the direction of assembly before the thread sealing compound has set. Any perceptible movement of any fin will be considered as failure of the assembly to comply with the drawing requirements.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.-Packaging, packing and marking shall be in accordance with the requirements of Level C in MIL-STD-1169.

5.2 Data card.-Data card information shall be in accordance with MIL-STD 1167 Ammunition Data Card.

6. NOTES

6.1 Visual examination qualification.-When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

CUSTODIAN:

Army - MU

Preparing Activity:

Army - MU

Project No. 1315 - A625

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.